

Shorter Checkouts

The US Department for homeland security is working on a system where fingerprinting and face recognition can be used to process frequent overseas travellers faster

How Humans See

The whole point of artificial intelligence is to try and create computing systems that operate more and more like humans and don't appear artificial at all. New research into the perception patterns of human beings has revealed certain strange and useful facts that are sure to have a significant impact on face recognition technologies.

Cues from the Face-blind: recent research into face blindness in humans has shown that the presence of emotional information in the face increases neural activity in the area of the brain called the FFA (Fusiform Face Area).

This proves that body and face-sensitive perceptions are dependent on emotional content and that neutral faces are more difficult to differentiate between. The FFA displays lower activation for neutral faces. Perhaps, then, our systems need to identify emotions in faces to make accurate assessments, just as humans do.

Remote control faces: a computer scientist in San Diego seems to have done just that. Jacob Whitehill, a computer science Ph.D. student from UC San Diego's Jacobs School of engineering has developed a technology for detecting facial expressions (part of an ongoing project at the school). Early last year, Whitehill successfully showed how his facial expressions could be turned into a remote control that speeds or slows video playback.

Robotic Eyes

Facial Expressions fall under the broad head of 'non-verbal communication'. At the Tokyo Institute of Technology, a team led by Yoichi Yamazaki have built and eye-robot which is nothing but a pair of 'eyeballs' which can convey a whole range of emotions. While the attempt is to make a robot more expressive, using eyes as



Digital billboards from Yahoo!

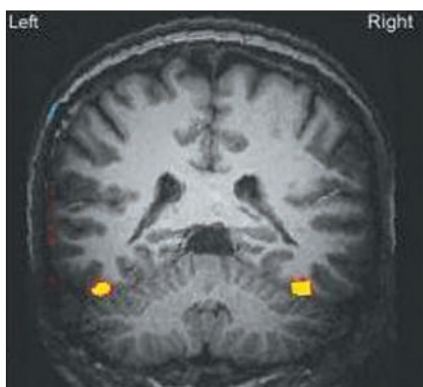
iWant porn

There is a flip side to face recognition as well, the iWant Porn application substitutes a user's face on porn clips and distributes them

we humans do, it stands to reason that if a machine can successfully emote, it can surely be made to identify human emotions as displayed on a human face before it.

Expressions in binary code

Jessica Dennis and Michael Ulrich, engineering students at the Rowan University College of Engineering, have spent 2008 working on developing a software program that has the capacity to read facial expressions. Using the JAFFE (Japanese Female Facial Expression) database, they arrived at four key measurements: the height and the width of the mouth and the height and width of the eyebrows. How wide the eyes were open and the jaw line were also considered as secondary measurements. The computer was then made to understand the expressions by translating them into binary code. After testing hundreds of photos, the duo determined that their software program had a 94.2 per cent accuracy for the emotions of surprise, fear and anger and an average of 76.5 per cent for all expressions.

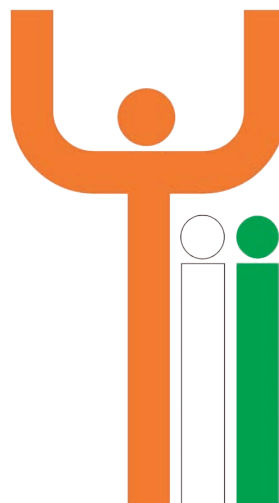


The Fusiform face area of the brain

Liar, Liar

Paul Ekman, a veteran in the field of study of human facial expressions, in studying people with mental disorders has successfully identified what he calls "microexpressions" – split second facial expressions – that betray the subject's emotional state. According to him, no one can fake a microexpression because they are inadvertent and slip out in spite of the subject's conscious attempt to repression. The web site www.mettonline.com was then set up by Ekman and his colleagues – a facility that anyone can log onto and train oneself in the art of picking out microexpressions that can show that someone is lying.

Tomorrow's tech



Digit has decided to compile a list of Young Indian Innovators who are changing our world through tech. Send your nominations to yii@thinkdigit.com. Visit www.thinkdigit.com/yii for more information

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This has now been taken forward by Mary Frank, a former student of Ekman's, presently teaching at the University of Buffalo, New York, with the help of computer scientists from the University of California. They have succeeded in automating Ekman's Facial Action Coding System (FACS), turning it into a technology called the Computer Expression Recognition Toolbox (CERT). This has tremendous potential for use in forensics, and if proved to be faultless, can be used not just on video footage of court trials but also in interviews, weddings and business contracts. The only point that ought to be perhaps considered here is the that now the machines are going to be better at reading faces than most of us humans are. What that portends for the human race, we leave you to contemplate.

Facing facts

To cut a long story short, research into facial recognition technologies is digging channels into territories that were deemed unconnected until a few years ago. The fact remains, however, that just as retina and fingerprint scanners can be fooled, faces can be altered (vide Michael Jackson) to outwit the most accurate algorithms. Even if Picasa's face-tagging software can distinguish between identical twins, there's a hacker born every minute and, in the long run, relying only on facial biometrics would be, well, myopic. It seems inevitable then, that a combination of several biometric technologies holds the key to a more secure identification system in the future. The latest in a list of biometric topics of interest are palm geometry and tracing the pattern of veins under the palm; the way a person walks (gait), the way a person bangs out text from his keyboard (keystroke dynamics), DNA and voice prints could be some possible combinations that could make face recognition more infallible. The right combination might just wipe that smile of the terrorist's face for good. **[d]**